Università della Svizzera Italiana

Faculty of Economics

Programming in Economics and Finance II

pyvar – A Python Package for Risk Management

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a.y. 2024/2025

Submission Date: June 13, 2025

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1 Introduction

Risk management is central to finance, yet many practitioners still rely on inefficient tools like spreadsheets. As programming becomes essential in financial workflows, there is a growing need for accessible, efficient solutions to compute Value-at-Risk (VaR) and other risk metrics.

We introduce `pyvar`, a Python package that automates the full VaR estimation pipeline—from portfolio creation to risk metrics, visualization and backtesting. Focused on equity portfolios and simple models, the package prioritizes speed, clarity, and ease of use—ideal for both professionals and retail investors seeking intuitive, modern risk tools.

2 Theoretical Foundation

This section outlines the mathematical foundations of our risk models. For simplicity, we group them into two categories: univariate and portfolio-level models. We begin with univariate models, which describe the return dynamics of a single asset or, equivalently, a portfolio treated as a single aggregated position. We then introduce portfolio-level models, which consider the joint distribution of multiple asset returns. These multivariate approaches capture dependencies between assets and allow for a more accurate assessment of portfolio risk. The section concludes with a presentation of the backtesting techniques used to evaluate the performance and reliability of the risk estimates produced by our models.

2.1 Basic VaR Models

Let $c \in (0, 1)$ denote the confidence level, and define $\alpha = 1 - c$ as the corresponding probability of a large loss. Value-at-Risk (VaR) at level α is defined as the worst loss L over a fixed horizon such that the probability of observing a larger loss is at most α:

$$\Pr(L > \mathrm{VaR}_\alpha) \leq \alpha.$$

Intuitively, this means we are confident that with probability c, losses will not exceed VaR_α over the given horizon.

By convention, VaR is expressed as a positive monetary amount. To simplify the notation in single-asset models we omit the scaling by invested wealth W and use percentage-based notation. To simplify the calculations we assume that W is strictly positive and that returns follow a zero-mean process.

We define the loss variable as $\ell_t = -r_t$. We start with models that assume constant volatility over time. The non-parametric (historical) and the parametric normal VaR are computed as:

$$\mathrm{VaR}_\alpha^{\text{hist}} = Q_\alpha(\ell_t), \qquad \mathrm{VaR}_\alpha^{\text{normal}} = z_\alpha \sigma,$$

where $Q_\alpha(\ell_t)$ is the empirical α-quantile of losses, and $z_\alpha = \Phi^{-1}(1 - \alpha)$ is the positive quantile of the standard normal distribution. For the Student-t VaR, the quantile is taken from the fitted Student-t distribution, estimated via maximum likelihood.

Expected Shortfall (ES), also known as Conditional VaR, is computed as:

$$\mathrm{ES}_\alpha^{\text{historical}} = \mathbb{E}\left[\ell_t \mid \ell_t \geq \mathrm{VaR}_\alpha\right], \quad \mathrm{ES}_\alpha^{\text{normal}} = \sigma \cdot \frac{\phi(z_\alpha)}{\alpha}, \quad \mathrm{ES}_\alpha^{t_\nu} = \sigma \cdot \frac{f_{t_\nu}(z_\alpha)}{\alpha} \cdot \frac{\nu + z_\alpha^2}{\nu - 1}$$

where ϕ is the standard normal density and f_{t_ν} is the density of the Student-t distribution with ν degrees of freedom.

The Cornish-Fisher expansion adjusts the normal quantile to account for higher-order moments of the return distribution — particularly skewness and kurtosis. This leads to a closed-form approximation of VaR:

$$\mathrm{VaR}_\alpha^{\mathrm{CF}} = -z^{\mathrm{CF}} \cdot \sigma,$$

where the adjusted quantile z^{CF} is given by:

$$z^{\mathrm{CF}} = z + \frac{1}{6}(z^2 - 1)s + \frac{1}{24}(z^3 - 3z)k - \frac{1}{36}(2z^3 - 5z)s^2,$$

with s and k representing the sample skewness and excess kurtosis, and z the standard normal quantile.

This expansion allows for asymmetry and fat tails in a parametric way, though it does not extend naturally to a consistent Expected Shortfall estimate.

The Hybrid or Exponentially Weighted Historical approach refines the historical approach by applying an exponentially decaying weighting scheme to past losses, giving more relevance to recent observations. The weights are defined as:

$$w_t = \alpha_0 \cdot \lambda^t, \quad \text{where } \alpha_0 = \frac{1 - \lambda}{\lambda(1 - \lambda^n)},$$

for $t = 0, 1, \ldots, n-1$, and $\lambda \in (0, 1)$ is a decay factor.

VaR and ES are computed from the weighted loss distribution as:

$$\mathrm{VaR}_\alpha^{\text{hybrid}} = Q_\alpha^{\mathrm{w}}(\ell_t), \qquad \mathrm{ES}_\alpha^{\text{hybrid}} = \frac{\sum_{t:\, \ell_t \geq \mathrm{VaR}_\alpha} w_t \ell_t}{\sum_{t:\, \ell_t \geq \mathrm{VaR}_\alpha} w_t}.$$

This approach partially captures time-varying risk and is particularly effective at reducing the ghost effect—where outdated extreme losses can cause abrupt drops in estimated risk.

2.2 Extreme Value Theory

Technically still a parametric model, EVT is more sophisticated than the methods discussed above. We implement it with the Peaks-Over-Threshold (POT) approach: once a high threshold u (e.g., the 99th-percentile loss) is chosen, a Generalized Pareto Distribution (GPD) is fitted to the excesses above u.

VaR and ES are then derived analytically from the fitted GPD parameters. This allows for more accurate estimation of rare, extreme losses. The formulas used are:

$$\mathrm{VaR}_\alpha = u + \frac{\beta}{\xi}\left[\left(\tfrac{n}{n_u}(1 - c)\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_\alpha = \frac{\mathrm{VaR}_\alpha + \beta - \xi u}{1 - \xi},$$

where ξ and β are the estimated shape and scale parameters, n is the sample size, and n_u is the number of exceedances.

2.3 Volatility Models

We have implemented the most popular univariate volatility models.

The GARCH(1,1) model is specified as:

$$\sigma_t^2 = \omega + a\, r_{t-1}^2 + b\, \sigma_{t-1}^2, \qquad r_t = \sigma_t z_t,$$

where z_t are i.i.d. standardized innovations. VaR is then computed as:

$$\mathrm{VaR}_t = \sigma_t z_\alpha.$$

Here, $z_\alpha = -Q_\alpha(z_t)$ is the positive α-quantile from the empirical distribution of the standardized innovations.

Future volatility can be forecasted analytically using GARCH(1,1).

The τ-step ahead forecast and cumulative T-step forecast are:

$$\begin{aligned}
\mathbb{E}[\sigma_{t+\tau}^2] &= \mathrm{VL} + (a+b)^\tau (\sigma_t^2 - \mathrm{VL}), \\
\mathbb{E}[\sigma_{t,T}^2] &= \mathrm{VL}\left(T - 1 - \frac{(a+b)(1-(a+b)^{T-1})}{1-(a+b)}\right) + \sigma_t^2\, \frac{1-(a+b)^T}{1-(a+b)},
\end{aligned}$$

which naturally allow multi-period VaR estimation.

We also support EGARCH (logs variance, capturing leverage), GJR-GARCH (adds an indicator for negative shocks), and APARCH (introduces a power term for flexible tail dynamics), all with maximum-likelihood estimation under Normal, Student-t, GED, or Skew-t innovations.

For simpler benchmarks we include ARCH(p) and two rolling estimators, Moving Average (MA) and Exponentially Weighted Moving Average (EWMA):

$$\sigma_t^{2,\mathrm{ARCH}} = \omega + \sum_{i=1}^{p} \alpha_i\, r_{t-i}^2, \qquad \sigma_t^{2,\mathrm{MA}} = \frac{1}{n}\sum_{i=1}^{n} r_{t-i}^2, \qquad \sigma_t^{2,\mathrm{EWMA}} = \lambda \sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2.$$

ES for volatility models becomes:

$$\mathrm{ES}_t = \sigma_t\, \mathbb{E}[-z_t \mid z_t < -z_\alpha].$$

All our volatility models use a semi-empirical approach for VaR computation: we first estimate the empirical distribution of standardized innovations z_t, then compute the required quantile z_α.

2.4 Time-Varying Correlation Models

We now introduce models designed to capture time-varying correlations at the portfolio level, extending beyond single assets. Specifically, we implement four approaches: moving average (MA), exponentially weighted moving average (EWMA), rolling principal component analysis (PCA), and Ledoit-Wolf (LW) shrinkage. These methods aim to provide more accurate and stable estimates of the conditional covariance matrix Σ_t, which governs portfolio-level risk.

Let $x_t = (x_{1,t}, \ldots, x_{N,t})^\top$ denote the vector of monetary positions, and let $w_t = x_t / \sum_{i=1}^N x_{i,t}$ be the corresponding vector of portfolio weights. We allow monetary positions in an asset to be negative, representing short positions. However, we assume the total portfolio value is strictly positive at all times.

Each method provides a different way to estimate the conditional covariance matrix. In the MA and EWMA models, Σ_t is estimated using a rolling sample or an exponentially weighted average of past return covariances, respectively.

In the PCA model, we compute a rolling sample covariance matrix and approximate it by retaining only the top eigencomponents, forming a low-rank matrix $\Sigma_t^{\text{PCA}} = V_t \Lambda_t V_t^\top$, where V_t contains the leading eigenvectors and Λ_t the corresponding eigenvalues.

For Ledoit-Wolf shrinkage, we estimate the covariance over a rolling window and compute a regularized matrix $\Sigma_t^{\text{LW}} = \delta_t F + (1 - \delta_t) S_t$, where S_t is the sample covariance matrix, F is a structured target (e.g., identity scaled by average variance), and $\delta_t \in [0, 1]$ is the shrinkage intensity estimated from the data.

In all cases, the 1-day VaR is computed as:

$$\text{VaR}_t = z_\alpha \cdot \sqrt{w_t^\top \Sigma_t w_t},$$

where Σ_t is the estimated variance-covariance matrix, obtained using one of the four methods described above: MA, EWMA, PCA, or Ledoit–Wolf shrinkage.

These models adapt naturally to both parametric and semi-empirical frameworks and assume a buy-and-hold strategy for portfolio positions. Expected Shortfall is computed using either the closed-form formula under normality or an empirical tail average of standardized residuals.

2.5 Factor Models

We implement the Sharpe model and Fama-French three factor models to estimate portfolio VaR and ES. These models use asset betas with respect to systematic risk factors, allowing us to infer portfolio-level risk from exposures to common sources of variation.

For the Sharpe Model, we assume that each asset's return r_i follows the linear factor structure:

$$r_i = \alpha_i + \beta_i r_m + \varepsilon_i,$$

where r_m is the return on the market portfolio, β_i is the asset's sensitivity to the market factor, α_i is the asset's abnormal return, and ε_i is a zero-mean idiosyncratic shock uncorrelated with r_m and other residuals. The market return r_m has variance σ_m^2, and $\text{Var}(\varepsilon_i) = \sigma_{\varepsilon_i}^2$. We support both a simplified setting with constant market volatility and a dynamic version where σ_m^2 is estimated via a GARCH model. Portfolio weights are assumed to be perfectly constant as well.

The total portfolio variance is:

$$\sigma_p^2 = \left(\sum_{i=1}^{N} w_i \beta_i\right)^2 \sigma_m^2 + \sum_{i=1}^{N} w_i^2 \sigma_{\varepsilon_i}^2.$$

VaR and ES are computed as follows under the normality assumption:

$$\text{VaR}_\alpha = z_\alpha \cdot \sigma_p, \qquad \text{ES}_\alpha = \sigma_p \cdot \frac{\phi(z_\alpha)}{\alpha}.$$

We also extend the previous model by considering three risk factors: market excess return (Mkt-RF), size (SMB), and value (HML). Each asset's excess return is modeled as:

$$r_i - r_f = \alpha_i + \beta_{i,1} \cdot \text{Mkt-RF} + \beta_{i,2} \cdot \text{SMB} + \beta_{i,3} \cdot \text{HML} + \varepsilon_i,$$

where r_f is the risk-free rate, and ε_i is the idiosyncratic error. We estimate factor loadings $\beta_{i,k}$ through OLS regression.

Let B be the $N \times 3$ matrix of estimated factor loadings, Σ_f the 3×3 sample covariance matrix of the factors, and Σ_ε the diagonal matrix of residual variances $\sigma_{\varepsilon_i}^2$. The total portfolio variance is:

$$\sigma_p^2 = w^\top (B \Sigma_f B^\top + \Sigma_\varepsilon) w.$$

Once the portfolio variance is known, we compute VaR and ES as we did for the single factor model.

2.6 Analytic VaR

We compute several Analytic VaR measures using the previously defined monetary position vector $x_t = (x_{1,t}, \ldots, x_{N,t})^\top$ and the covariance matrix Σ.

The quantile z_α corresponds to the desired tail probability level and is always derived from the standard normal distribution, as the underlying assumption is that asset returns are normally distributed.

The asset-normal parametric portfolio VaR is defined as:

$$\text{VaR}_t = z_\alpha \cdot \sqrt{x_t^\top \Sigma x_t} \cdot \sqrt{h}.$$

Assuming all asset returns are perfectly correlated ($\rho = 1$) and there are no short positions, the undiversified portfolio VaR is simply the sum of the individual asset VaRs:

$$\text{UVaR}_t = \sum_{i=1}^{N} \text{VaR}_{i,t}.$$

The marginal VaR, representing the sensitivity of portfolio VaR to a small change in position $x_{i,t}$, is given by:

$$\Delta\text{VaR}_{i,t} = \text{VaR}_t \cdot \frac{(\Sigma x_t)_i}{x_t^\top \Sigma x_t}.$$

It measures the change in total VaR from adding one extra unit of asset i.

The component VaR is:

$$\text{CVaR}_{i,t} = x_{i,t} \cdot \Delta\text{VaR}_{i,t},$$

which captures the absolute contribution of asset i to total portfolio VaR.

The relative component VaR is:

$$\text{RCVaR}_{i,t} = \frac{\text{CVaR}_{i,t}}{\text{VaR}_t},$$

interpreted as the percentage share of total VaR attributable to asset i.

For a vector a representing changes in the portfolio allocation, the incremental VaR is:

$$\text{IVaR}_t = \Delta\text{VaR}_t^\top \cdot a,$$

which estimates the change in total VaR resulting from shifting the portfolio by a.

Each of these VaR metrics can be complemented with a corresponding Expected Shortfall (ES), computed under the normality assumption.

2.7 Options

Delta-Normal approximations provide fast and analytically tractable methods for estimating the VaR of portfolios containing European options. These methods rely on the Black-Scholes model and are particularly effective for short-term risk under normal market conditions. However, when portfolios include more complex or nonlinear instruments, simulation-based approaches—which will be introduced next—offer a more robust alternative.

The Black-Scholes model gives the fair value of a European call or put option as:

$$c_t = N(d_1)S_t - N(d_2)Ke^{-r(T-t)}, \qquad p_t = N(-d_2)Ke^{-r(T-t)} - N(-d_1)S_t,$$

where

$$d_1 = \frac{\ln(S_t/K) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Here, S_t is the spot price, K the strike price, $T-t$ the time to maturity in years, r the continuously compounded risk-free rate, σ the annualized volatility of returns, and $N(\cdot)$ the standard normal cumulative distribution.

The option's delta, which measures sensitivity to the underlying asset, is given by $\Delta = N(d_1)$ for a call and $\Delta = N(d_1) - 1$ for a put. Using this, a first-order approximation of the VaR_α for a single option position with q contracts, each on N units of the underlying asset, is:

$$\text{VaR}_\alpha = \alpha \cdot |\Delta| \cdot S \cdot \sigma \cdot q \cdot N,$$

where α is the quantile of the standard normal distribution corresponding to the confidence level.

For a portfolio of options on h underlying assets, we compute the dollar delta exposure for each asset as $x_i = \Delta_i S_i$, and estimate VaR_α using the variance-covariance matrix Σ of asset returns:

$$\text{VaR}_\alpha = \alpha \cdot \sqrt{\mathbf{x}^\top \Sigma \mathbf{x}}.$$

These formulas provide a fast approximation of market risk, assuming that portfolio values are linearly sensitive to underlying prices and that asset returns are normally distributed.

2.8 Simulation Methods

Simulation methods are powerful tools for estimating VaR and ES, particularly in portfolios with non-linear instruments such as options.

We begin with the one-day Monte Carlo approach under the assumption of multivariate normality. Future returns are simulated as:

$$\mu = \mathbb{E}[r_t], \quad \Sigma = \text{Cov}(r_t), \quad L = \text{Cholesky}(\Sigma),$$

$$r^{(i)} = \mu + Lz^{(i)}, \quad z^{(i)} \sim \mathcal{N}(0, I),$$

$$S^{(i)} = S_0 \circ \exp(r^{(i)}),$$

where $\circ$ denotes element-wise multiplication. Portfolio profit and loss is calculated as:

$$\Delta P^{(i)} = w^\top (S^{(i)} - S_0) + \sum_{j=1}^{N_{\text{opt}}} q_j \left(C(S_j^{(i)}, \tau_j') - C_{j,0} \right),$$

where q_j is the number of option contracts, $C(\cdot)$ is the Black-Scholes option price, and $\tau_j' = \max(T_j - \frac{1}{252}, 0)$ is the adjusted time to maturity.

We define simulated portfolio losses as $\ell^{(i)} = -\Delta P^{(i)}$. From the empirical distribution $\{\ell^{(i)}\}_{i=1}^N$, where N is the number of simulations, we compute:

$$\text{VaR}_\alpha = Q_\alpha(\ell), \qquad \text{ES}_\alpha = \mathbb{E}[\ell \mid \ell \geq \text{VaR}_\alpha].$$

This method is extended to multiple days (equity-only), where asset paths are iteratively simulated and used to compute terminal portfolio losses. These multi-step simulations serve as a forward-looking, non-analytical forecast of risk.

To better capture return asymmetry and regime switching, we implement a Gaussian Mixture Model (GMM) extension. In the multivariate case:

$$r^{(i)} \sim \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(\mu_k, \Sigma_k),$$

where K is the number of Gaussian components (typically between 2 and 4), and π_k are the mixture weights. Prices are propagated like before.

We also support a univariate GMM version, which fits a mixture to log-returns of a single asset. As with Monte Carlo, both univariate and multivariate GMMs can be extended to multiday horizons, although only for equity portfolios.

We also implement univariate GARCH-based simulations. A GARCH(1,1) model is fitted to historical log-returns, and future volatility paths are simulated under zero drift ($\mu = 0$) to avoid artificial upward trending in asset paths. Two innovation types are supported: standard normal shocks and standardized Student-t shocks for heavier tails. Prices are again obtained using the same exponential propagation.

As a non-parametric alternative, we implement Historical Simulation and its bootstrap variant. Historical (or resampled) log-returns are applied to current prices using the same method as before. The bootstrap variant resamples with replacement to improve robustness in small-sample settings.

The Weighted Historical Simulation variant improves responsiveness to recent market behavior by applying exponentially decaying weights.

This matches the approach used in hybrid VaR and ES, applying exponential weights to emphasize recent losses.

Finally, Filtered Historical Simulation combines GARCH modeling with empirical innovations. A GARCH model is fitted to each asset, and standardized residuals z_t are resampled and rescaled using the most recent conditional volatility $\hat{\sigma}_T$:

$$r^{(i)} = \hat{\sigma}_T \cdot z^{(i)},$$

with returns propagated to prices as before. This approach captures time-varying volatility while remaining non-parametric in distributional assumptions.

Across all methods, except for the WHS case, Expected Shortfall is computed as:

$$\mathrm{ES}_\alpha = \mathbb{E}[\ell \mid \ell \geq \mathrm{VaR}_\alpha].$$

All simulations assume static portfolio weights and do not explicitly model volatility or interest rate risk beyond what is embedded in the return process. No backtesting functionality is included.

2.9 Backtesting

Backtesting assesses whether the observed losses are consistent with the VaR predictions by checking the frequency and structure of violations (exceptions). The objective is to test whether the model is correctly calibrated.

We implement three standard tests.

The Kupiec test (unconditional coverage) evaluates if the number of observed exceptions N over T days matches the expected number under the model. If p is the failure probability, then under the null

N follows a binomial distribution. The likelihood ratio test statistic is:

$$\mathrm{LR}_{\mathrm{uc}} = -2\left\{\ln\left[(1-p)^{T-N}p^N\right] - \ln\left[\left(1-\frac{N}{T}\right)^{T-N}\left(\frac{N}{T}\right)^N\right]\right\}, \quad \mathrm{LR}_{\mathrm{uc}} \sim \chi_1^2.$$

The Christoffersen test (conditional coverage) tests the independence of exceptions by modeling their dynamics as a first-order Markov chain. It compares the likelihoods of transition counts under the null (independence) and alternative. The statistic is:

$$\mathrm{LR}_{\mathrm{c}} = -2(\ln L_0 - \ln L_1), \quad \mathrm{LR}_{\mathrm{c}} \sim \chi_1^2,$$

where L_0 is the likelihood under independence and L_1 under the alternative.

The joint test combines both:

$$\mathrm{LR} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{c}} \sim \chi_2^2.$$

Rejecting the null in any of the tests suggests that the VaR model is either miscalibrated (too many or too few violations) or fails to capture time dependence in risk (e.g., volatility clustering).

Our back-testing framework covers every method except the simulation models. Each simulation delivers only a single-day (or multiple day) forecast; to back-test them we would need to rerun a full rolling simulation at every step—a task that is both logically more involved and computationally heavy—so it lies outside the present scope.

3 Results and Applications

Virtually all applications and results of the `pyvar` package can be found in the `examples` folder of the GitHub repository. A series of Jupyter notebooks illustrate typical use cases, all following a consistent setup: the creation of a simple portfolio followed by the use of the package's functions to calculate various risk metrics and visualize them effectively.

Each notebook includes additional insights and brief theoretical refreshers to support understanding and contextualize the results.

4 AI Acknowledgement

We used ChatGPT-4o extensively for coding support—mainly to generate pseudocode or early drafts, which we refined and corrected. GitHub Copilot also assisted with code completion. For theory extraction (e.g., formulas from books/slides), Notebook LM was particularly helpful. ChatGPT-o3 supported us with theoretical clarifications, while ChatGPT-4o was also used for LaTeX help and documenting our code with clean docstrings.

5 Link to GitHub Repository

The full codebase is available on GitHub: `https://github.com/Alessandro-Dodon/pyvar`

The repository includes:

- This documentation
- Example notebooks demonstrating usage
- Plots illustrating the results
- The full source code of the `pyvar` package

6 References

The following textbooks and lecture notes constitute the main references for this project:

- John C. Hull, *Risk Management and Financial Institutions*, 2023, John Wiley & Sons.
- Peter Christoffersen, *Elements of Financial Risk Management* (2nd Edition), Academic Press, 2011.
- Philippe Jorion, *Value at Risk*, McGraw-Hill, 2006.
- Andrea Resti and Andrea Sironi, *Risk Management and Shareholders' Value in Banking*, John Wiley & Sons, 2007.
- John C. Hull, *Options, Futures, and Other Derivatives*, latest edition, Pearson.
- Kevin Sheppard, *Financial Econometrics Notes*, University of Oxford, February 2, 2021.

Additional insights and context were drawn from the following MSc in Quantitative Finance courses at Università della Svizzera Italiana, Lugano:

- *Risk Management*, taught by Prof. A. Plazzi
- *Derivatives and Financial Econometrics*, taught by Prof. L. Mancini

We also acknowledge the valuable content and feedback from:

- Francesco Violante, *Market Risk Management*, IÉSEG School of Management, Paris – kindly suggested by Andrea Mercier and Farah Saoudi, who also provided insightful comments on several modules of the package.